SEOP Platform Architecture

Seven pipeline layers · Cross-cutting services · Data foundation · BURR evaluation · CMMI-5 process integration

Pipeline (7 Layers)

Feedback Loop

L0 · Seed Studio (M01)

Seed contract · CQs · Scope · Upper-ontology commitment · Versioned registry · Human approval

Output: immutable seed ID referenced by all downstream layers

L1 · Ingestion & Context (M15 · M02)

Multimodal ingestion · Real-time + batch · Schema introspection · BURR signals (normalization, FK, N:M, entropy, prompt-length)

Simulation requirement analysis · Draft CQs · Provenance per chunk · DQ quarantine (M16)

Output: structured seed candidates + routing signals

L2 · Seed Extraction (M03 · M04 · M05)

NER · RE · GLiNER · LLM prompts · Candidate classes, properties, instances · Provenance + confidence

Statistical grounding (DB columns, API payloads, ranges) · Hallucination rejection · Disambiguation & canonicalization

Output: grounded, canonical seed candidates

L3 · Synthesis (M06 · M07 · M08 · M41 · M44)

Upper ontology alignment (BFO/DOLCE) · Domain synthesis · Restrictions · Disjointness · Decision chain per axiom

Prompt Strategy Engine (4 tiers) · Mapping Format Generator (R2RML / D2RQ / RML / Direct)

Serialization: Turtle, RDF/XML, OWL Manchester, SHACL/ShEx drafts

Output: candidate TBox + mappings

L4 · Validation (M09 · M10 · M12)

Syntactic · Structural (SHACL/ShEx) · Logical (HermiT/Pellet/ELK) · Quality (OOPS!/OntoClean)

Standards conformance (14+ vocabularies, ISO 21838/29002) · Human Review Gate · Dual review

Output: validated TBox + validation report + human decisions

L4.5 · ABox & CQ Testing (M11)

Instance extraction · Entity resolution · SPARQL CQ execution · Coverage & gap report

Output: populated knowledge graph + functional proof via CQ pass rate

L5 · Evolution (M13 · M14)

Simulation deployment (DT / ABM) · Error monitoring · LLM patch proposals · Impact analysis

Regression testing (BURR) · Human approval · Versioned release · Rollback · Continuous learning loop

Output: evolved ontology version + patch audit trail

Cross-Cutting Services

Data Foundation (M15–M20)

M15 Ingestion · M16 Quality · M17 Dictionary · M18 Lifecycle

M19 Governance · M20 Metadata & Ontology

BURR signals: normalization, FK, N:M, entropy, naming drift

DQ gates · Canonical naming normalizer · Lineage

Approved terms ⟡ M17 ⟡ M40 lexical scoring

Metadata catalog ⟡ M20 ⟡ M39 benchmark artifacts

CMMI-5: business glossary at Level 5 includes ontology structures

Enterprise Foundation (M26–M31)

M26 RBAC & IAM · M27 Provenance & Audit · M28 Registry & Versioning

M29 Evaluation Harness · M30 Security & Compliance · M31 LLM Orchestrator

Provenance: PROV-O decision chain per element · prompt tier · model · reviewer

Registry: SemVer · diff · rollback · BURR scores archived per version

Security: SOC 2 · ISO 27001 · GDPR · HIPAA · PII redaction

LLM Orchestrator: prompt versioning · cost control · fewshot injection

BURR Evaluation Suite (M39–M46)

M39 Benchmark Harness · M40 Dual-Metric Engine · M41 Prompt Strategy

M42 Model Routing · M43 Hybrid Ensemble · M44 Mapping Format

M45 Scenario Library · M46 Statistical Rigor Service

Dual metrics: lexical F1 + mapping F1 per element (C/R/A)

Hallucination detection: high lexical / low mapping

Bootstrap CI · Nemenyi · CD diagrams · significance gate

R2RML default · D2RQ · RML · Direct Mapping

Regression gate for M14 patch approval

CMMI-5 Process Areas (M32–M38)

M32 OPP · M33 QPM · M34 CAR · M35 DAR

M36 CM · M37 PPQA · M38 MA

MA (M38): BURR metric stream is primary measurement source

QPM (M33): SPC control charts on BURR F1

CAR (M34): root cause on BURR score drops

DAR (M35): formal model/prompt promotion decisions

CM (M36): scenarios and mappings are baselines

Enterprise Consumers

Agentic AI runtime: SPARQL / GraphQL · ontology-driven tool selection

Digital Twin platform: semantic simulation · KPI drift monitoring

BI & Analytics: dashboards · BURR scorecards · statistical reports

External APIs: SPARQL 1.1 · GraphQL · REST · webhooks · Kafka

Standards-conformant exports: R2RML / D2RQ / RML

RBAC enforced at gateway (M26)

Standards · Compliance · Provenance

BFO · DOLCE · ISO/IEC 21838 · ISO 29002 · R2RML · D2RQ · RML · Schema.org · DCAT · DCTERMS · SKOS · PROV-O · GeoSPARQL · WGS84 · SSN · SOSA · SAREF · FIBO

CMMI Level 5 (Continuous Representation) · SOC 2 · ISO 27001 · GDPR · HIPAA · SPARQL 1.1 · OWL 2 · SHACL · ShEx · 100% provenance coverage